

NORTH PLATTE, NE, USA

WMO: 725620

Lat: 41.1328N

Lon: 100.7000W

Elev: 848

StdP: 91.54

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 24023

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.8	-17.1	-23.5	0.5	-18.9	-20.8	0.7	-16.1	13.8	1.0	12.0	2.1	2.9	310	0.613

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.3	35.7	21.3	33.9	21.5	32.1	21.1	24.0	31.1	23.1	30.3	22.3	29.5	5.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.9	18.4	27.4	21.0	17.4	26.7	20.1	16.4	25.9	76.5	31.2	73.0	30.4	69.8	29.5	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.2	10.7	9.0	-25.4	38.6	2.7	2.2	-27.3	40.2	-28.9	41.5	-30.5	42.8	-32.4	44.4
			-25.6	25.9	2.7	1.1	-27.5	26.7	-29.1	27.3	-30.6	27.9	-32.5	28.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-3.3	-1.8	3.9	8.8	14.5	20.7	24.1	22.7	17.8	9.9	2.8	-2.8
	DBStd	10.94	6.08	6.34	6.06	5.22	4.76	4.00	3.16	3.19	4.87	5.10	5.38	5.64
	HDD10.0	1727	412	331	202	82	13	1	0	0	3	64	221	398
	HDD18.3	3591	670	564	448	288	137	21	2	4	67	264	467	657
	CDD10.0	1663	0	1	12	45	152	322	436	393	238	61	4	0
	CDD18.3	487	0	0	0	1	18	92	181	140	53	3	0	0
	CDH23.3	5834	0	1	19	68	320	1136	2034	1448	711	94	4	0
	CDH26.7	2605	0	0	3	14	102	482	1024	659	298	23	0	0
Wind	WSAvg	4.0	3.7	4.0	4.5	5.1	4.5	4.1	3.6	3.4	3.8	4.0	3.7	3.6
Precipitation	PrecAvg	507	10	13	27	55	84	89	77	54	41	34	13	11
	PrecMax	752	28	50	76	151	203	222	179	160	153	121	49	65
	PrecMin	256	0	0	0	2	19	11	4	2	3	1	0	0
	PrecStd	102	7	12	20	34	44	43	41	35	36	27	12	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.0	20.4	25.9	28.8	32.7	36.5	38.9	37.2	35.2	29.7	23.3	16.5
		MCWB	7.2	9.0	12.4	14.3	18.6	20.9	21.4	21.2	19.5	15.8	11.4	7.6
	2%	DB	12.6	15.6	21.9	25.2	29.5	33.9	36.1	34.7	32.8	25.9	19.2	12.3
		MCWB	5.2	6.7	10.7	13.7	18.2	21.4	21.9	21.5	19.5	14.7	9.6	5.3
	5%	DB	9.3	12.0	18.5	22.4	26.9	31.6	34.1	32.6	30.3	23.0	15.9	9.3
		MCWB	3.6	5.4	9.3	12.4	17.1	20.7	22.2	21.5	19.0	13.4	8.0	3.7
	10%	DB	6.1	8.4	15.1	19.2	24.4	29.4	32.3	30.5	27.6	19.9	12.4	6.1
		MCWB	1.9	3.4	7.7	11.0	15.7	19.9	22.1	21.3	17.8	12.2	6.5	1.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.7	9.7	14.4	16.4	20.9	24.1	25.5	24.7	22.6	18.1	12.4	8.3
		MCDB	15.9	18.3	20.9	25.7	28.8	31.3	32.4	31.3	30.3	25.4	19.9	15.6
	2%	WB	5.6	7.3	12.0	14.5	19.3	22.8	24.2	23.6	21.0	16.2	10.4	5.7
		MCDB	12.2	14.5	20.1	22.3	26.9	30.3	31.5	30.5	29.1	23.1	17.9	12.0
	5%	WB	3.9	5.5	10.1	13.0	18.0	21.7	23.5	22.8	20.0	14.6	8.7	3.8
		MCDB	8.8	11.6	17.3	21.0	25.0	29.2	30.7	29.7	27.6	21.0	15.5	8.9
	10%	WB	2.1	3.6	8.2	11.6	16.6	20.7	22.7	21.9	18.9	12.9	6.7	1.9
		MCDB	5.8	8.2	15.1	18.4	23.0	27.9	30.0	28.3	25.5	18.9	12.5	5.8
Mean Daily Temperature Range	5% DB	MDBR	14.5	14.4	15.9	14.9	14.3	14.4	14.3	14.2	15.9	15.9	15.8	14.5
		MCDBR	19.7	20.7	22.5	21.3	18.4	17.2	17.4	17.1	18.8	21.4	21.9	19.5
		MCWBR	12.9	12.7	12.5	11.0	8.9	7.3	6.7	6.8	7.9	11.0	12.8	12.9
	5% WB	MCDBR	18.7	19.9	20.1	18.4	16.1	15.1	15.0	14.8	16.1	17.8	20.5	18.9
		MCWBR	12.6	12.4	11.5	9.9	8.2	7.3	6.9	7.0	7.4	9.8	12.3	12.7
Clear-Sky Solar Irradiance	taub	0.261	0.269	0.294	0.322	0.352	0.366	0.387	0.379	0.346	0.309	0.272	0.262	
	taud	2.543	2.521	2.476	2.417	2.411	2.403	2.388	2.393	2.447	2.519	2.547	2.522	
	Ebn at Noon	919	959	963	950	923	907	883	882	892	895	894	885	
	Edh at Noon	69	83	98	112	115	117	117	113	100	83	69	66	
All-Sky Solar Radiation	RadAvg	2.29	3.18	4.31	5.33	6.20	7.11	7.03	6.11	4.95	3.51	2.52	1.98	
	RadStd	0.19	0.21	0.34	0.30	0.43	0.44	0.26	0.28	0.25	0.34	0.16	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page